



Population-based clustering of co-occurring social determinants: An application of unsupervised machine learning

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ABSTRACT

Purpose: This study aimed to develop a cluster-based measure of multiple co-occurring social determinants of health by applying unsupervised machine learning to a population-based cohort, offering a data-driven approach to organize complex social exposures.

Methods: Unsupervised clustering was applied to a population-based cohort of Ontario respondents to six-cycles of the Canadian Community Health Survey (2001–2012) linked to the Canadian census and vital statistics data. Clusters were evaluated using internal metrics, visualization techniques, descriptive analysis and theoretical considerations to determine the optimal number of clusters. Sensitivity analyses were integrated across the iterative clustering process. Premature mortality rates were generated assess validity.

Results: Optimal clustering solutions included 4-clusters and 6-clusters. Both cluster solutions revealed distinct social typologies. The 6-cluster solution offered greater granularity and theoretical interpretability. The 4-cluster solution showed greater heterogeneity within certain marginalized groups. Premature mortality rates differed meaningfully across clusters, supporting the clustering approach in capturing risk associated with social exposure.

Conclusions: Unsupervised machine learning methods identified meaningful population subgroups reflecting complex patterns of social exposures. This approach offers a flexible, data-driven method for characterizing social exposures that can be considered alongside theoretical frameworks and used for equity monitoring, intervention planning and policy development.

Introduction

The social determinants of health (SDOH) [1] are fundamental drivers of health and health inequity [2]. They operate upstream of the health system and encompass a wide range of economic, environmental, political, social and cultural factors [3,4]. SDOH act across multiple levels (e.g., individual, community, structural) and include systemic

factors like income inequality and unemployment which shape differential exposure to health risks [5]. These overlapping exposures contribute to the persistent health inequities observed within and between populations [6].

In observational health research, SDOH are often analyzed individually, either as exposures, effect modifiers, or covariates [7]. These approaches may underrepresent the complexity of social exposures, as

Abbreviations: CCHS, Canadian Community Health Survey; ML, Machine Learning; PCA, Principal Component Analysis; RPDB, Ontario Registered Persons' Database; SDOH, Social Determinants of Health; t-SNE, t-Distributed Stochastic Neighbor Embedding; UMAP, Uniform Manifold Approximation and Projection.

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individuals are simultaneously influenced by multiple co-occurring SDOH. Traditional statistical methods struggle to capture this complexity due to parametric assumptions, multicollinearity, and reliance on assumptions of linearity and additivity, which may not align well with the inherently correlated nature of social exposures (e.g., education and income) [8].

There is a growing body of literature in which summary measures of multiple social exposures are generated e.g., polysocial risk scores (PsRS) [9–12], however these methods typically involve reducing the selected social exposures into a single measure (e.g., composite score). While this approach can be straightforward and easy to interpret, it may be an oversimplification and obscure heterogeneity [13]. There remains a lack of methods to systematically organize and analyze multiple co-occurring SDOH at the population level.

Partitioning clustering identifies distinct groups, by minimizing within-cluster variability and maximizing inter-cluster differences [14, 15]. Latent class analysis [16], a probabilistic clustering approach commonly implemented in applied health research, typically requires assumptions such as local independence and can become computationally demanding as dimensionality increases [17], while PsRS approaches reduce multiple social determinants to a single composite score. Partitioning-based clustering approaches, such as k-prototypes, offer an alternative analytical strategy that groups observations using an additive, per-variable dissimilarity structure designed for mixed data types. This approach holds different structural assumptions than alternatives (e.g., LCA-style latent variable models), reflecting a distinction between model-based inference and distance-based pattern discovery in settings characterised by highly correlated social exposures. This study addresses a gap in population health research by applying scalable and transparent analytic approaches to jointly characterize co-occurring social exposures in a large population-based dataset. Specifically, a clustering-based analytic pipeline is applied to mixed categorical and continuous social exposure data derived from linked population-based data, which remain the primary source of SDOH information in population health research. The aim of this work is to develop a cluster-based measure that captures the multidimensional and co-occurring nature of social exposures.

By preserving the multidimensional structure of the data, unsupervised clustering may identify population subgroups with similar social exposure profiles, which can be used descriptively to examine population-level patterning of social exposure or as exposure groupings in etiologic analyses, supporting more context-specific public health interventions.

Methods

Data sources

This study used six-cycles of the Canadian Community Health Survey (CCHS), a cross-sectional survey conducted by Statistics Canada that collects self-reported demographics and health data. With the use of survey weights, the data are representative of 98% of the non-institutionalized Canadian population aged 12 and older. Notably, it excludes people living on First Nations Reserves, Crown Lands, in institutions, in remote regions and full-time members of the Canadian forces. The CCHS survey methodology has been described extensively elsewhere [18].

CCHS respondents were linked to administrative databases (1992–2022) held at ICES (previously the Institute for Clinical Evaluative Sciences) which captures data from Ontario's single-payer insurance system [19]. Linkages included the Ontario Registered Persons' Database (RPDB), a population registry of Ontario residents who have been eligible for the province's single payer health insurance program since 1992; the Ontario Registrar General's death file (1999–2022) and area-level measures from the Canadian census (1991, 2001–2011) based on postal code of residence at time of CCHS interview. Postal codes were

mapped to census geography at the dissemination area level using Statistics Canada's Postal CodeOM Conversion File Plus (PCCF+). Dissemination areas are the smallest unit for which population characteristics from the Canadian Census are reported and have populations ranging from 400 to 700 individuals. These datasets were linked using unique encoded identifiers and analyzed at ICES [19].

This study was approved by the Research Ethics Board at the University of Toronto.

Study cohort

Ontario respondents to the 2000–2012 cycles of the CCHS who consented to administrative linkage were included in this study. Individuals were excluded if they resided outside of Ontario at the time of interview; were under the age of 25 or 65 years or older at the time of interview; appeared in multiple CCHS cycles (only first occurrence was used); had an invalid birth date, death date or sex recorded; or had a date of last contact with the health system that preceded their interview date.

Measures

Individual-level measures

Individual-level measures were identified from the respondent's CCHS interview and reflect self-reported, single point-in-time assessment. Measures reflect five domains of social exposure [12] and include: Population Group (immigration status, racialized (visible minority), speaks French or English); Economic Circumstances (household annual income relative to median, worked in the past 12 months, had a job last week, social assistance reliant, food insecurity over past 12 months); Education (respondent, household); Community and Social Context (cohabitation, community belonging, living arrangements); Material Circumstances (home ownership, household crowding) [12]. Rurality (yes/no), derived from the CCHS was also included. In addition, the Rurality Index of Ontario, a measure which reflects distance to health services, was used to further capture rurality [20].

Area-level measures

Census-based measures reflected the domains of the Ontario Marginalization Index (ON-Marg) (age and labour force, households and dwellings, material resources, and racialized and newcomer populations) [21]. Additional census-based measures reflecting education, skills, and training deprivation were also incorporated [22]. Canadian census-based measures at the dissemination-area level were assigned to respondents using postal code of residence at time of CCHS interview.

Premature mortality

Premature mortality reflects deaths that occur before an expected age, highlighting preventable health losses and the effectiveness of health systems in addressing avoidable risks. Premature mortality was defined as death prior to age 75, in line with previous literature in this population [23] and the accepted definition in Canada [24]. Study members were followed up for premature mortality, within 5-years and 10-years from the CCHS interview date. Mortality was confirmed through linkage with vital statistics information from the RPDB.

Data pre-processing

A total of 16 categorical and 28 continuous measures were considered for the cluster analysis. To manage outliers and ensure comparability across measures with different scales, continuous variables were winsorized [25] at the 1st and 99th percentiles, where appropriate, prior to standardization [26]. Pearson's correlation coefficient was used to assess relationships among continuous variable pairs while Cramér's V coefficient was used for associations between categorical variables.

Variables with a correlation coefficient greater than 0.9 were removed to minimize multicollinearity. Principal component analysis (PCA) was applied to the continuous variables to reduce variable redundancy prior to clustering, with the resulting component scores used as clustering inputs. Specifically, components explaining 90% of the total variance [15,27,28] were retained ensuring that continuous measures contributed comparably across the SDOH domains. Further methodological details are provided in [Supplementary Table 2](#).

K-prototype clustering

Several clustering algorithms were initially explored across a range of algorithmic families, including partitioning, density-based and model-based approaches to assess feasibility and alignment with the structure and scale of the data. To further guide the selection of algorithm, an OPTICS reachability plot was generated to explore the underlying structure of the data and assess the appropriateness of a partitioning based approach [29].

The k-prototypes algorithm, a hybrid of k-means and k-modes, was selected for clustering as it was explicitly designed to accommodate mixed continuous and categorical data [30], a common feature of routinely collected population survey data on social determinants of health. In addition, k-prototypes scales efficiently to large sample sizes and produces cluster assignments which are readily interpretable, aligning with the applied and descriptive aims of this study.

K-Prototypes represents one partitioning-based choice among several possible alternatives, rather than a universally preferred clustering method. Its use is appropriate in settings where mixed data types, large sample sizes, and sets of correlated indicators are present, and where analytic transparency and interpretability are primary considerations. Clustering was implemented using the *kmodes* Python package (v0.12.2, 2024) [31]. For each execution, the weighting parameter was set using the default estimation, which provides a heuristic scaling between numeric and categorical dissimilarities without requiring manual tuning, but does not eliminate sensitivity to variable scaling [30,32]. To identify the optimal number of clusters, ten iterations were conducted for each *k* value ranging from 2 to 12 using a 10% random sample of the preprocessed data owing to the computational demands associated with the large sample size and analytic grid runtime constraints (4-h limit, 7CPUs). Key metrics (cost, silhouette score, Davies-Bouldin index, and cohesion) [33] were averaged and plotted, and cluster characteristics were reviewed by subject matter experts.

Using the final variable set, sex-stratified clusters were generated on the full dataset, for both a data-driven optimal *k*, and subject-matter informed *k*. Descriptive statistics, along with a narrative summary, were used to characterize each cluster. A Sankey plot was created to visualize the transitions between cluster solutions.

The CCHS sampling weights were not incorporated in the clustering procedure as the k-prototype algorithm does not directly accommodate sampling weights. Weights were applied after clustering to produce population weighted descriptive summaries [34].

Sensitivity analyses

Clustering was repeated under alternative scenarios retaining 80% variance and with no dimensionality reduction, to assess the impact of feature compression on cluster structure. Additional sensitivity analysis was then conducted at the optimal number of clusters to evaluate the contribution of individual continuous variable. Clusters were regenerated with each variable sequentially removed, and performance was assessed using cost and silhouette scores. To assess the stability of cluster assignments, sensitivity analyses were conducted using a 20% random subsample of the data. Multiple models were generated varying random seed values, binary social exposure indicators and missing data flags. Agreement in cluster assignments across models was evaluated using Cramér's V. All analyses were sex-stratified.

Cluster validation

Cluster validation relies on internal metrics, visualization, and external comparisons to assess whether clusters reflect meaningful real-world patterns. Cluster metrics were averaged across ten repeated iterations. Visualization was performed using PCA, UMAP and t-SNE, each reducing the data to two dimensions and labeling points by cluster assignment [27,35,36]. These methods offer complementary views of the data structure, with PCA capturing variance, UMAP preserving the local and global data structure [36] and t-SNE highlighting local relationships [35].

To further validate the clusters, we compared the clusters to an external population health outcome (premature mortality). Crude and age-standardised premature mortality rates were calculated for each sex-stratified cluster, applying CCHS sampling weights and using the 1991 Canadian census population as reference [24].

Full analytic details are provided in the [Supplementary material](#). Data preparation and analyses were performed at ICES using SAS (Version 8.3; SAS Institute Inc., Cary, NC, USA) and Python (Version 3.6.15; Python Software Foundation, Wilmington, DE, USA). Reporting of AI-related methods was guided by the APPRAISE-AI framework [37].

Results

Study population

A total of 62,368 females and 53,523 male CCHS were included in this study ([Fig. 1](#)). Sex-stratified cohort characteristics are presented in [Table 1](#). While the distribution of characteristics was largely similar across sexes, some differences were observed. A higher proportion of females were in the lowest income group compared to males, and females were also more likely to have not worked in the past 12 months, had food insecurity and relied on social assistance compared to male respondents. There were minimal differences between the sexes across area-level measures.

Data pre-processing

Highly correlated variables (Pearson's *r* or Cramér's *V* > 0.9) were removed to reduce redundancy. The OPTICS reachability plot suggested a partitioning structure, supporting the use of a non-density-based clustering approach.

K-prototype clustering

The k-prototypes algorithm was applied across *k* = 2–12 with PCA-based dimensionality reduction (90%, 80% variance explained, and no dimension reduction). A 4-cluster solution emerged as optimal based on

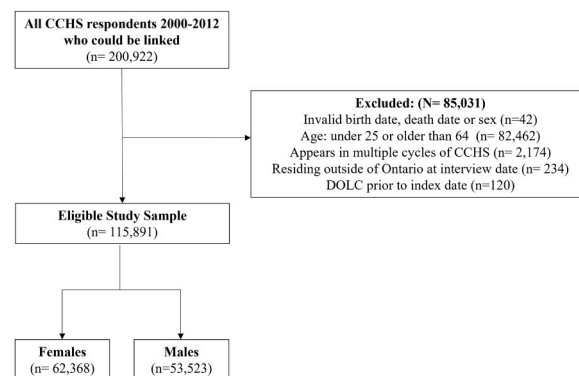


Fig. 1. Eligibility criteria for study participants, Canadian Community Health Survey, 2001 to 2012.

Table 1

Characteristics of the Ontario CCHS study population (age 25–64) at interview date, by sex, 2000–2011.

	Females n = 62,368		Males n = 53,523	
	n	Weighted %	n	Weighted %
Age, mean ± SD	43.7 ± 18.6		43.9 ± 17.9	
25–34	14,365	23.8 (23.3, 24.4)	11,478	23.5 (22.9, 24.1)
35–44	15,997	29.1 (28.4, 29.7)	14,837	29.5 (28.9, 30.2)
45–54	15,143	27.0 (26.3, 27.7)	13,384	26.5 (25.8, 27.1)
55–64	16,863	20.1 (19.6, 20.6)	13,824	20.5 (20.0, 21.1)
Population Group				
Immigration status				
Canadian Born	40,366	68.9 (68.1, 69.7)	35,191	70.6 (69.7, 71.4)
Immigrated lt 10 yrs ago	2701	11.2 (10.6, 11.8)	2072	9.7 (9.1, 10.3)
Immigrated mt 10 yrs ago	6016	19.8 (19.1, 20.5)	5276	19.7 (19.0, 20.5)
Cultural Racial Origin				
White	55,626	77.2 (77.2, 77.2)	47,852	78.1 (78.1, 78.1)
Racialized	6743	22.8 (22.8, 22.8)	5672	21.9 (21.9, 21.9)
Speaks an official CDN language				
Yes	61,515	96.5 (96.4, 96.6)	53,038	97.7 (97.5, 97.7)
No	853	3.5 (3.4, 3.6)	485	2.3 (2.3, 2.5)
Economic Circumstances				
Household income (relative to median)				
Group 1: Lowest - 0 to < 0.5 median income	7765	11.8 (11.6, 11.9)	5114	9.0 (8.9, 9.1)
Group 2: 0.5 to < 1	18,072	26.1 (26.0, 26.3)	13,208	23.2 (23.1, 23.4)
Group 3: 1 to < 1.5	20,809	34.1 (33.8, 34.2)	18,975	36.9 (36.7, 37.1)
Group 4: 1.5 to < 2	6274	10.5 (10.4, 10.6)	5756	11.0 (11.0, 11.1)
Group 5: Highest > 2	9448	17.6 (17.5, 17.7)	10,471	19.9 (19.8, 19.9)
Worked in the past 12 months				
Yes	47,358	77.5 (77.5, 77.5)	46,251	90.3 (90.3, 90.3)
No	15,010	22.5 (22.5, 22.5)	7272	9.7 (9.7, 9.7)
Had a job last week				
Yes	43,804	71.9 (71.9, 71.9)	43,658	85.7 (85.7, 85.7)
No	18,564	28.1 (28.1, 28.1)	9865	14.3 (14.3, 14.3)
Social Assistance Main source of Income				
No	56,984	93.7 (93.5, 93.7)	49,931	95.6 (95.5, 95.6)
Yes	5384	6.3 (6.3, 6.5)	3592	4.4 (4.4, 4.5)
Experienced some food insecurity in the past 12 months				
Food insecure	6770	10.3 (10.2, 10.4)	4307	7.2 (7.1, 7.3)
Not food insecure	55,598	89.7 (89.6, 89.8)	49,217	92.8 (92.7, 92.9)
Education				
Education - Respondent				
LT Secondary	7132	10.2 (10.1, 10.3)	7318	11.3 (11.3, 11.4)
Secondary	12,030	18.8 (18.8, 18.9)	9905	17.5 (17.4, 17.5)
MT Secondary	43,205	71.0 (71.0, 71.1)	36,301	71.2 (71.1, 71.3)
Education - household				
LT Secondary	4543	5.4 (4.6, 6.1)	3931	5.1 (4.8, 5.4)
Secondary	8096	11.1 (11.0, 11.2)	7244	11.3 (11.2, 11.6)
MT Secondary	49,730	83.5 (82.8, 84.3)	42,349	83.6 (83.0, 84.0)
Community and social context				
Partner status				
Married or Common law	30,724	65.7 (64.9, 66.4)	25,067	63.0 (62.2, 63.8)
Widowed, Separated, Divorced	6052	9.0 (8.6, 9.4)	3995	5.5 (5.2, 5.9)
Single, Never Married	12,307	25.3 (24.7, 26.0)	13,477	31.5 (30.7, 32.2)
Community Belonging				
Very Strong	10,770	16.3 (16.2, 16.4)	9035	15.9 (15.9, 16.0)
Some Strength	30,712	48.0 (47.9, 48.1)	25,589	46.7 (46.6, 46.8)
Some Weakness	15,183	25.0 (24.9, 25.1)	13,847	27.1 (27.0, 27.1)
Very Weak	5703	10.7 (10.6, 10.7)	5053	10.3 (10.2, 10.5)
Lives alone				
Yes	12,541	9.9 (9.9, 9.9)	12,109	11.3 (11.3, 11.3)
No	49,827	90.1 (90.1, 90.1)	41,414	88.7 (88.7, 88.7)
Material Circumstances				
Home Ownership				
Resides in owned home	47,889	76.3 (76.3, 76.4)	41,560	76.7 (76.7, 76.8)
Residence is not owned	14,479	23.7 (23.6, 23.7)	11,963	23.3 (23.2, 23.3)
2 Person/Bdrm	2551	6.4 (6.2, 6.5)	2202	6.6 (6.5, 6.7)
Household Crowding				
LT 1 Person/Bdrm	29,419	36.0 (35.5, 36.7)	25,276	35.7 (35.2, 36.2)
1 Person/Bdrm	29,495	55.3 (54.6, 55.8)	25,179	55.3 (54.9, 55.8)
2 Person/Bdrm	2551	6.4 (6.2, 6.5)	2202	6.6 (6.5, 6.7)
3 + Person/Bdrm or no bedrooms	904	2.3 (2.0, 2.6)	866	2.3 (2.1, 2.5)
Rurality				
Not Rural	49,151	89.3 (89.3, 89.3)	41,957	88.6 (88.6, 88.6)
Rural	13,217	10.7 (10.7, 10.7)	11,566	11.4 (11.4, 11.4)
Rurality Index Ontario* mean ± SD	9.8 ± 21.5		10.2 ± 21.9	

* Rurality Index of Ontario (RIO) reflects distance to health services and is a proxy for health care access.

clustering metrics. A feature ablation study showed minimal impact on cost and silhouette scores, supporting retention of all continuous features.

While metric evaluation identified a 4-cluster solution as optimal, additional solutions were reviewed by subject matter experts in social epidemiology. The 6-cluster solution was selected as it captured greater granularity and better reflected population heterogeneity, while maintaining interpretability and theoretical relevance. Visualization of the clustered data using both linear (PCA) and non-linear (t-SNE, UMAP) dimensionality reduction methods provided complementary views of data structure, showing broadly separable yet partially overlapping clusters and comparable multidimensional patterns across sexes (Fig. 2; $k = 4$, Supplementary Fig. 5).

A narrative summary of the clusters is provided in Box 1, with a heatmap of the 6-cluster solution in Fig. 3 (Supplementary Tables 3–4). Clustering was performed separately for males and females; however, as comparable patterns were observed across sexes, a single integrated summary is presented for conciseness.

In the 4-cluster solution, substantial heterogeneity was observed within socially disadvantaged groups. As displayed in the Sankey plot (Supplementary Fig. 7), expanding to six clusters, an empirically reasonable alternative, revealed two distinct subgroups predominantly composed of racialized individuals: Cluster C, characterized by economic disadvantage, and Cluster D, reflecting relative socioeconomic security.

Narrative description of clusters ($k = 6$)

Six clusters (A–F) were identified, each reflecting distinct socio-demographic profiles. The cluster descriptions integrate individual-level characteristics with neighborhood-level census measures, summarized by cluster membership. Area-level characteristics reflect the average neighborhood context of individuals in each cluster.

Cluster A – predominantly white, canadian born, economically marginalized

Cluster A comprises primarily white, Canadian-born individuals with

lower educational attainment and higher economic marginalization. On average, these individuals mostly reside in neighbourhoods with low immigrant and racialized populations, moderate population density (3500/sq km), and high reliance on social assistance. These areas also show elevated rates of one-parent families and material deprivation related to housing and household resources.

Cluster B – economically and socially marginalized

This cluster represents low-income individuals living in dense urban areas, characterized by financial hardship, crowded housing and low home ownership. On average, these individuals tend to live in areas with older populations (+20% aged 65+), high dependency ratios and limited labour force participation. While more diverse than Cluster A, at the neighbourhood level this cluster remains less diverse than the population average. High residential turnover, a predominance of rental housing and a high proportion of individuals living alone are also notable.

Cluster C – immigrant-dense, economically disadvantaged urbanites

Cluster C comprises a highly racialized, immigrant-dense population facing multiple layers of disadvantage. Individuals in this cluster typically live in extremely dense (17,000/sq km), rental-heavy and transient urban areas. These neighbourhoods are marked by high deprivation with approximately 35% living below the low-income cut-off, unemployment rates exceeding 10%, and around 60% of the population self-identifying as racialized.

Cluster D – stable immigrants with socioeconomic security

Cluster D predominantly includes racialized and immigrant individuals who are economically secure. This cluster has high rates of home ownership, larger households and high community belonging. On average, this cluster resides in neighbourhoods with a large racialized and newcomer population, like Cluster C, but without the same levels of material deprivation and structural disadvantage.

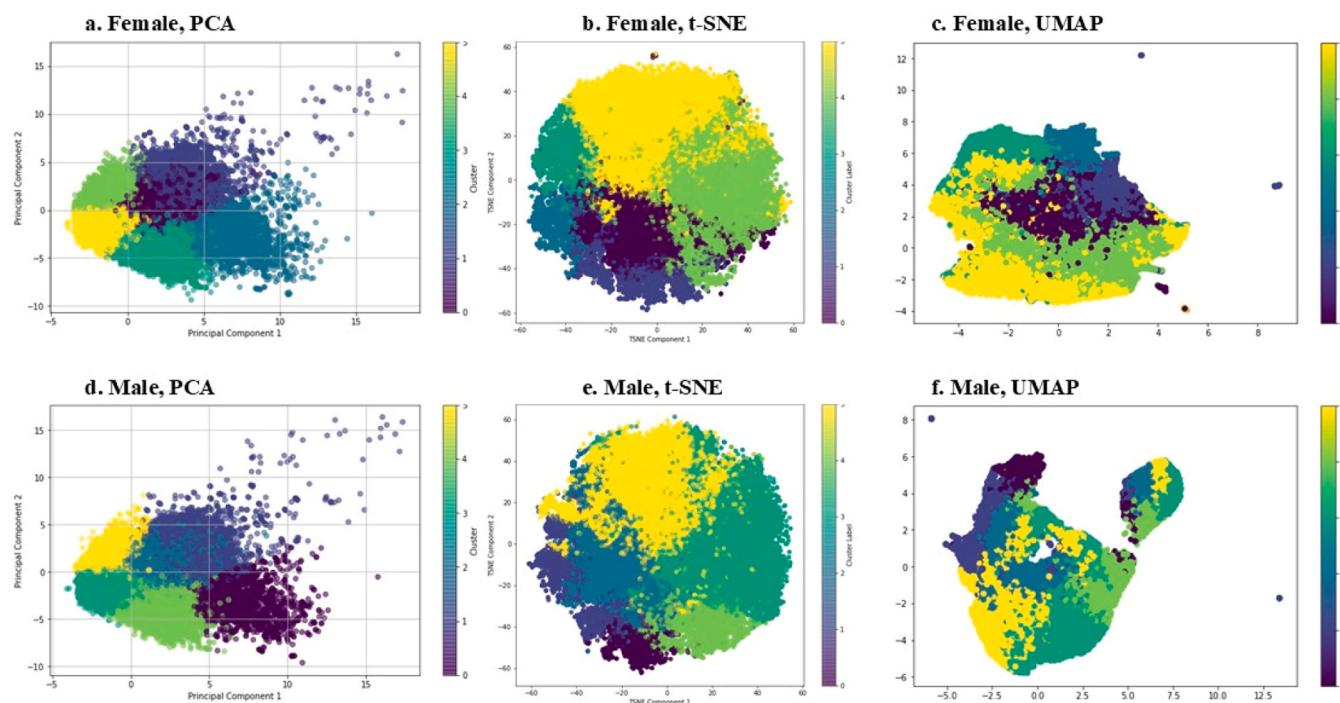


Fig. 2. Visualisation of sex-stratified clustering patterns across three dimensionality reduction methods, 6-cluster solution.

Box 1

Narrative cluster summary.

Metric informed cluster solution ($k = 4$)		Alternative cluster solution ($k = 6$)	
Cluster	Description	Cluster	Description
Cluster 1 (Females: 8.7%; Males: 8.5%)	Predominantly White, Socially Disadvantaged	Cluster A (Females: 11.8%; Males: 11.9%)	Canadian Born, Predominantly White, Economically Marginalized
Cluster 2 (Females: 16.7%; Males: 16.5%)	Predominantly Immigrant & Racialized, Socially Disadvantaged	Cluster B (Females: 7.0%; Males: 7.5%)	Economic and Social Marginalization
Cluster 3 (Females: 38.4%; Males: 38.1%)	Socially Advantaged (High Education, Employment, Income)	Cluster C (Females: 9.8%; Males: 8.7%)	Immigrant-Dense, Economically Disadvantaged Urbanites
Cluster 4 (Females: 36.2%; Males: 36.9%)	Older, Rural, Canadian-born, White, Socially Stable	Cluster D (Females: 17.3%; Males: 17.0%)	Stable Immigrants with Socioeconomic Security
		Cluster E (Females: 18.0%; Males: 19.4%)	White Rural Homeowners with Socioeconomic Stability
		Cluster F (Females: 36.1%; Males: 35.4%)	Predominantly White, Socioeconomically Secure

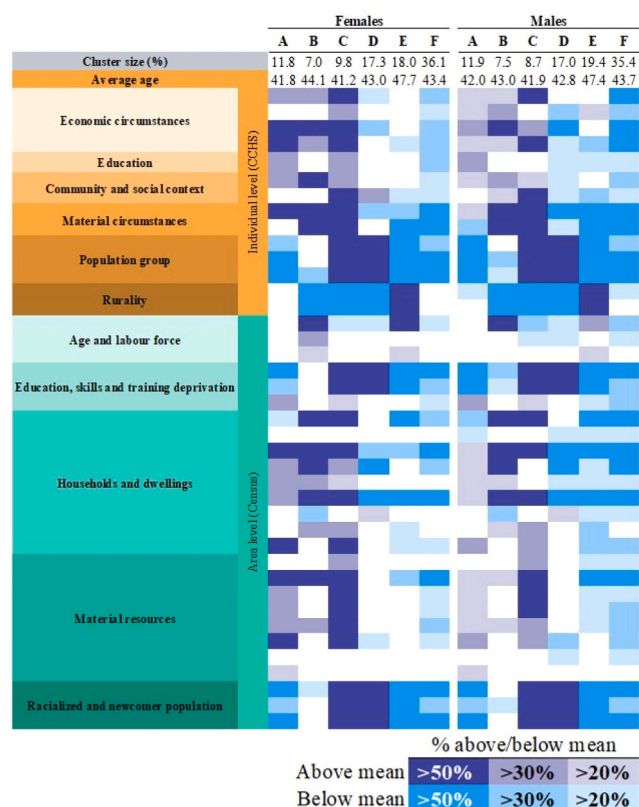


Fig. 3. Sex-stratified heatmap illustrating relative social exposure patterns across cluster solutions.

Cluster E – white rural homeowners with socioeconomic stability

Cluster E consists primarily of white (95%), Canadian-born (87%), adults living in low-density rural or small-town areas. The average age of individuals in this cluster is slightly higher than the population average. On average, individuals in this cluster reside in neighbourhoods with high home ownership, low poverty and low population density.

Cluster F – predominantly white, socioeconomically secure

Cluster F is the largest cluster, reflecting approximately 35% of the

sample. It includes socioeconomically secure, predominantly white (89%), mostly Canadian Born (78%), adults with high rates of home ownership. On average, these individuals live in affluent neighbourhoods with high home ownership, low population density (2500/sq km) and minimal marginalization across multiple indicators.

Sensitivity analyses

Sensitivity analysis showed minimal differences across dimension reduction methods and no change in performance metrics when individual features were removed, indicating robust clustering results. Additional analyses assessing cluster membership stability were conducted separately for $k = 6$ solutions on a 20% subsample. Across all comparisons, cluster assignments demonstrated moderate to strong agreement (Cramér's $V > 0.65$). Given this consistent agreement, no further sensitivity analyses were conducted.

Premature mortality across clusters

Crude and age-standardized premature mortality rates (per 1000) for the 6-cluster solution are highlighted in Table 2 (4-cluster solution, Supplemental Table 7). Among females, Cluster A showed highest premature mortality within both 5- and 10-years of the CCHS interview, followed by Clusters B and C, while Cluster D had the lowest. Among males, Clusters A and C had the highest standardized premature mortality rates at both follow-up periods, while Clusters D and F had the lowest across both timeframes. These patterns indicate a social gradient in mortality that differed modestly by sex, with distinct cluster profiles associated with elevated risk in each sex.

Discussion

This study applied unsupervised clustering to a population-based survey cohort with linked individual and area-level SDOH data to identify groups of multiple co-occurring social determinants. Two cluster solutions were developed: a 4-cluster solution guided by internal metrics and a 6-cluster solution informed by both metrics and theory. While both produced distinct groupings, the 4-cluster solution showed heterogeneity within disadvantaged groups, particularly among racialized and immigrant populations. A Sankey plot illustrated how the 6-cluster solution further differentiated these groups. The immigrant-dense cluster identified in the 4-cluster model separated into two subgroups in the 6-cluster solution, one which reflected socioeconomic disadvantage (Cluster C), and another indicating socioeconomic security

Table 2

Sex-stratified comparison of 5-year and 10-year premature mortality rates and standardized premature mortality rates (per 1000) following CCHS interview date across 6-cluster solution (Unweighted: females n = 62,368; males n = 53,523).

	5 - Year		10 - Year	
	Crude rate	Standardized rate (95% CI)	Crude rate	Standardized rate (95% CI)
Female				
Overall	2.09	1.18 (1.09,1.28)	2.40	1.36 (1.29,1.43)
Cluster A	2.63	1.72 (1.38,2.12)	3.31	2.24 (1.95,2.56)
Cluster B	2.74	1.44 (1.08,1.88)	3.14	1.65 (1.37,1.96)
Cluster C	2.39	1.65 (1.28,2.1)	2.06	1.42 (1.17,1.71)
Cluster D	1.30	0.79 (0.61,1.01)	1.50	0.88 (0.75,1.04)
Cluster E	2.85	1.17 (0.98,1.38)	3.48	1.51 (1.35,1.68)
Cluster F	1.71	1.00 (0.86,1.16)	1.96	1.17 (1.06,1.29)
Male				
Overall	3.31	1.87 (1.75,1.99)	3.63	2.06 (1.97,2.15)
Cluster A	4.15	2.70 (2.27,3.19)	4.55	2.98 (2.65,3.34)
Cluster B	3.29	1.86 (1.45,2.36)	4.29	2.47 (2.12,2.86)
Cluster C	5.45	3.63 (3.03,4.31)	4.83	3.38 (2.95,3.86)
Cluster D	2.06	1.32 (1.07,1.61)	2.34	1.50 (1.31,1.71)
Cluster E	4.45	1.88 (1.65,2.15)	4.83	2.10 (1.91,2.29)
Cluster F	2.50	1.46 (1.28,1.65)	2.86	1.68 (1.54,1.82)

Premature mortality (<age 74).

Rates per 1000.

Standardized rates: 1991 Canadian census as reference population.

(Cluster D). These subgroups showed corresponding divergence in premature mortality, highlighting the complexity of social disadvantage and the value of a theoretically informed, data-driven approach.

Differences in premature mortality across clusters were consistent with established social gradients [23,38], confirming that the clusters captured meaningful variation in social risk. This work aligns with prior studies demonstrating the utility of cluster-based methods for identifying complex social and health patterns [32,39–41] and may inform future research on population segmentation and policy tailored to social exposure typologies.

While gradients were evident in both sexes, the specific clusters associated with higher mortality differed, suggesting that the social and structural profiles shaping health risk may vary by gendered social contexts. Notably, Cluster D, which was characterized by socioeconomic security within largely racialized and immigrant populations, showed lower premature mortality among females than males, suggesting gendered differences in the protective effects of social advantage. These findings reflected multidimensional social profiles rather than single determinants, highlighting the potential of this approach to represent complex social contexts relevant to health and health equity.

Unsupervised ML is well-suited for analyzing complex, multidimensional data, making it ideal for studying multiple co-occurring SDOH. Unlike singular indices, clustering preserves the multidimensionality of social exposures, enabling a more nuanced understanding [10,12,38]. Importantly, there is no single “correct” number of clusters [32,42]; the choice should be guided by the intended application and informed by both relevant metrics and theoretical expertise [43]. In this study, the six-cluster solution provided conceptual richness [4] and granularity, while the four-cluster solution may be preferable in contexts prioritizing simplicity or interpretability (e.g., surveillance). Recognizing this flexibility, with full transparency, is essential for applying unsupervised methods in public health, where trade-offs between granularity and utility must be carefully considered.

While the specific clustering results offer insights into social complexity at the population level, a key contribution lies in demonstrating the feasibility and value of this methodological approach [44]. By leveraging multiple sources of existing data (i.e., administrative, census and survey data) this approach captures the complexity of SDOH at the population level. As data sources expand to include unstructured data such as clinical notes via large language models, geospatial and environmental data, the methodological foundation demonstrated here can support increasingly sophisticated analyses of social complexity.

Limitations

This study used a representative population-based sample with linked individual and area-level SDOH data, which is a key strength of this study. The application of unsupervised clustering identified population groupings of social exposures, moving beyond linear models of cumulative risk. However, several limitations should be noted. First, individual-level data were self-reported, and assessed at a single time point. While SDOH vary across the life course, the inclusion of indicators such as education and employment may partially account for life-stage exposures. Second, the clustering results may be influenced by unmeasured variables, such as environmental exposures (e.g., access to greenspace, distance to transportation), which were not included in the analysis. Third, the derived clusters reflect data-driven patterns among respondents and certain important population subgroups were excluded from the CCHS, including residents of remote communities; institutionalized individuals (e.g., correctional facilities and nursing homes); persons living on reserves and Crown lands [45]. Although survey weights were applied during descriptive analyses to support population representativeness, the clustering procedure itself did not incorporate these weights as most unsupervised algorithms, including k-prototype, do not directly accommodate complex survey designs. This approach, which focuses on pattern identification rather than population estimation, is consistent with prior applications of k-prototype to survey data [34]. Additionally, clustering solutions can depend on algorithm choice and implementation; alternative approaches may yield different cluster solutions. While k-prototypes was selected as a reasonable and interpretable partitioning-based approach, it represents one of several possible analytic choices rather than a universally preferred method. Alternative implementation decisions (e.g., variable representation, dimensionality reduction, distance specification and weighting parameters) may lead to different cluster solutions, that may be preferable in other applications or across data sources. These considerations highlight the importance of aligning clustering methods with analytical goals and data characteristics. Finally, the clusters were generated on working age adults (25–64 years) and may not be generalizable to other age groups.

Conclusions

This study demonstrates the utility of a cluster-based approach for identifying meaningful population subgroups that capture the real-world complexity of social disadvantage. The resulting clusters offer a

multidimensional view of social exposures, offering a new approach to more deeply study health outcomes and carry out equity monitoring [46]. Future research is needed to evaluate the broader applicability of the social exposure clustering approach, particularly for informing targeted interventions and policies addressing the social determinants of health.

CRediT authorship contribution statement

Arjumand Siddiqi: Writing – review & editing, Supervision. **Peter M. Smith:** Writing – review & editing, Supervision. **Ingrid Giesinger:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Emmalin Buajitti:** Writing – review & editing, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Laura C. Rosella:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Rahul G. Krishnan:** Writing – review & editing, Supervision. **Gemma Postill:** Writing – review & editing, Methodology.

Ethical statement

Research Ethics Board approval for this study was obtained through the University of Toronto Health Sciences Research Ethics Board (Protocol #: 00041762).

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Conflict of Interest Disclosure statement

The authors have no conflicts of interests to declare.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.annepidem.2026.110062](https://doi.org/10.1016/j.annepidem.2026.110062).

Data availability statement

The dataset from this study is held securely in coded form at ICES. While legal data sharing agreements between ICES and data providers (e.g., healthcare organizations and government) prohibit ICES from making the dataset publicly available, access may be granted to those who meet pre-specified criteria for confidential access, available at www.ices.on.ca/DAS (email: das@ices.on.ca). The full dataset creation plan and underlying analytic code are available from the authors upon request, understanding that the computer programs may rely upon coding templates or macros that are unique to ICES and are therefore either inaccessible or may require modification.

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